

A Brief Exposition to Type Theory

(Bachelor's Thesis Presentation)

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A foundational system is a framework that allows mathematicians to build mathematical objects and theories in.

Zermelo-Frankel Set Theory is not complete by itself, since it lives in the language of *first-order logic*.

A **Foundational System** refers to both parts, the logical system and the infrastructure to express mathematical objects.

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Concretely, a Foundation System consists of

- A **Logical/Deductive system** specifying *rules of inference* that dictate which inferences are allowed from existing context.
- An **Axiomatic system**, a schematic collection of *axioms*, statements in the language of the deductive system, taken as true by assumption.

Usually in set-theoretic foundations, the former is *first-order logic* and the latter is *ZF or ZFC Set Theory*. Type Theory combines these two layers into the same framework.

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(Intensional) Martin-Löf Type Theory

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Some Basic Rules

- Types are primitive objects.
- Type A may be inhabited by a *term/instance*, denoted $a : A$.
- A *judgement* is an inference derived from existing context by *rules of inference*.
- Judgements are either of the form:

- $a : A$

- $a \equiv b : A$ or simply $a \equiv b$

where ' $\equiv$ ' stands for *judgemental equality*, an internal equality.

- There is also *propositional equality* denoted ' $=$ ', a weaker notion corresponding to equality as mathematical objects.

(Intensional) Martin-Löf Type Theory

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Logic in Type Theory

- Types also function as logical objects, in particular, they can simulate propositions.
- A term $a : A$ where A is a proposition is a *proof* of A .
- Logic emerges through rules of inference given by Type Theory.
- Logical quantifiers such as $\wedge, \vee, \implies, \forall, \exists$ correspond to certain constructions over types.
- Type families $E : A \rightarrow U$ such that $E(a)$ are propositions for $a : A$ correspond to properties of terms of A .
- Type Theory's logic is constructive!

Universes

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Type Theory postulates a chain of cumulative universes à la Russell:

$$\mathcal{U}_0, \mathcal{U}_1, \mathcal{U}_2, \mathcal{U}_3, \dots$$

satisfying:

- $\mathcal{U}_i : \mathcal{U}_{i+1}$ for each i
- If $A : \mathcal{U}_i$, then $A : \mathcal{U}_{i+1}$

Each type is a term of some universe. Each universe is also a term of universes above it.

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There is a general schema for constructing new types, requiring:

- **Formation rules** to introduce an inductive type into context.
- **Constructors/introduction rules** to introduce terms of the type.
- **Elimination rules** to consume terms of a type.
(Usually by a *recursion* or an *induction* principle which define (dependent) functions on the type)
- **Computation rules** to specify action of elimination on the constructors.

These rules specify the behaviour of an inductive type by dictating the inference rules for the type in the foundational system.

Examples

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Natural Numbers Type ($\mathbb{N}$) The natural numbers are an inductive type, given by the constructors

■ $\text{zero} : \mathbb{N}$

■ $\text{succ} : \mathbb{N} \rightarrow \mathbb{N}$

To prove a property $E : \mathbb{N} \rightarrow \mathcal{U}$, it suffices to prove $E(\text{zero})$ and for all $n : \mathbb{N}$, $E(n) \rightarrow E(\text{succ}(n))$.

This is the usual *principle of mathematical induction*!

Type Formers

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Type Former	Notation	Logical Analogue
Function Type	$A \rightarrow B$	$A \implies B$
Dependent Function Type	$\prod_{(x:A)} B(x)$	$\forall x \in A, B(x)$
Dependent Pair Type	$\sum_{(x:A)} B(x)$	$\exists x \in A, B(x)$
Product Type	$A \times B$	$A \wedge B$
Coproduct Type	0	$A \vee B$
Empty Type	0	False
Unit Type	1	True

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- Equalities are encoded as an inductive type.
- For any $A : \mathcal{U}$ and $x, y : A$, proofs of equality $x = y$ are terms of $\text{Id}_A(x, y)$. (Also denoted $x =_A y$)
- There is a constructor $\text{refl}_a : a =_A a$.
- But one can now treat $(x =_A y)$ as a mathematical object to predicate over.
- One can have multiple, nonequivalent proofs of equality.
- Can have iterated identity types $\text{Id}_{\text{Id}(x,y)}(p, q)$ etc.
- Bookeeping these nonequivalences ad infinitum captures structure.

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- Bookeeping these nonequivalences ad infinitum captures structure.

Remark. An equality $p : A =_{\mathcal{U}} B$ in a universe gives a transport $p_* : E(A) \rightarrow E(B)$ for any $E : A \rightarrow \mathcal{U}$.

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Homotopy Type Theory

(Weak) ∞ -Groupoid

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A (weak) ∞ -groupoid is a collection of:

- 0-cells called points
- 1-cells between two points
- ...
- $(n + 1)$ -cells between two n -cells

following

- Concatenation and identity leading to associativity and unit up to higher dimensional cells.
- All $(n + 1)$ cells being invertible up to $(n + 2)$ cells.

The Homotopy Interpretation

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In HoTT, types in HoTT can be interpreted as either:

- topological spaces upto weak homotopy equivalence,
- (weak) ∞ -groupoids from higher dimensional category theory.

One can establish an equivalence between these interpretations via the *Homotopy Hypothesis*.

Homotopy Hypothesis (Grothendieck)

A space X is determined upto weak homotopy equivalence by its fundamental groupoid $\Pi_{\infty}(X)$.

Identity Types as Path Spaces

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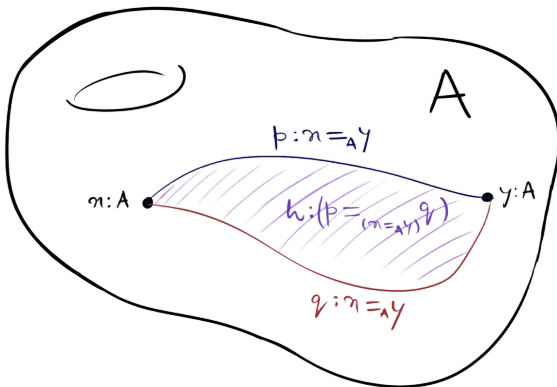
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Identity Types have a homotopical interpretation in HoTT:



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Identity Types have a homotopical interpretation in HoTT:

Types	$\leftrightarrow$	Spaces
Terms $x : A$	$\leftrightarrow$	Points x of A
$p : \text{Id}_A(x, y)$	$\leftrightarrow$	Paths from x to y
$h : \text{Id}_{\text{Id}_A(x, y)}(p, q)$	$\leftrightarrow$	Homotopies between parallel paths $p, q : x =_A y$
...	$\leftrightarrow$	...

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Consider the groups

$$(\mathbb{Z}/2\mathbb{Z}, +) \quad \text{and} \quad (\{\pm 1\}, \cdot)$$

Obviously isomorphic, so must share properties and theorems proven in group theory.

But no obvious way to translate theorems from one to another!

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In practice, mathematicians tend to informally identify isomorphic objects, and extend properties by abuse of notation.
To capture this formally, Voevodsky proposed the Univalence axiom:

Axiom (Univalence)

Let $A, B : \mathcal{U}$, then we have an equivalence

$$(A \simeq B) \simeq (A =_{\mathcal{U}} B)$$

- Given equivalent types, one may transport theorems and proofs along the path provided by univalence!
- Univalence formalizes this mathematical practice.

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Despite providing a strong interpretive shift, HoTT fails to be completely satisfactory due to the following

- Lacks good computational properties
- Univalence is an axiom, so is computationally troublesome
- *Computational canonicity* and *normalization* are not known to hold

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- Lacks good computational properties
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- *Computational canonicity* and *normalization* are not known to hold

Alternatives such as **Cubical Type Theory** are partially satisfactory but further work is needed.

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